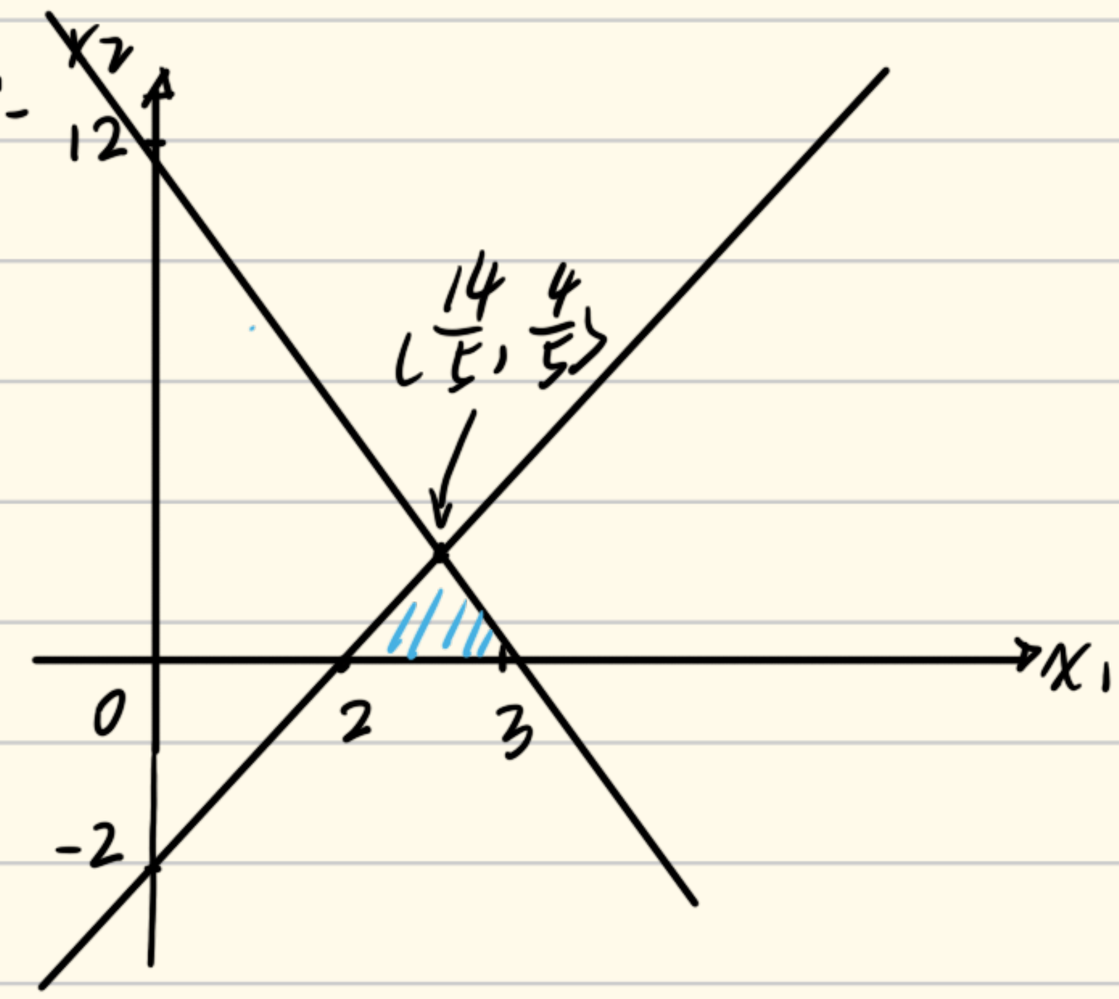
T1 $\because 4x_1 + x_2 \le 12,\ x_1 - x_2 \ge 2,\ x_1 \ge 0,\ x_2 \ge 0$

$\Rightarrow x_2 \le -4x_1 + 12,\ x_2 \le x_1 - 2,\ x_1 \ge 0,\ x_2 \ge 0$.

express this problem geometrically:



the blue region is the feasible region.

consider maximize

$z = c_1 x_1 + 2x_2$

$\Rightarrow x_2 = -\frac{c_1}{2} x_1 + \frac{z}{2}$

the vertex in boundary: $(\frac{14}{5}, \frac{4}{5}), (2,0), (3,0)$

$\because z \ge 0 \quad \therefore c_1 \ge 0$.

when $c_1 = 8$, $x_2 = -\frac{c_1}{2} x_1 + \frac{z}{2}$ has the same slope of $x_2 = -4x_1 + 12$, so the optimal solution for (x_1, x_2) is: $\{(x_1, x_2) \mid x_2 = -4x_1 + 12,\ x_1 \in [\frac{14}{5}, 3]\}$, $z = 12$.

when $c_1 \in (0, 8)$, geometrically the optimal solution for (x_1, x_2) is $(\frac{14}{5}, \frac{4}{5})$

when $c_1 \in (8, +\infty)$, geometrically the optimal solution for (x_1, x_2) is $(3, 0)$

when $c_1 = -2$, $x_2 = -\frac{c_1}{2} x_1 + \frac{z}{2}$ has the same slope of $x_2 = x_1 - 2$, so the optimal solution for (x_1, x_2) is $\{(x_1, x_2) \mid x_2 = x_1 - 2\ (x_1 \in [2, \frac{14}{5}])\}$, $z = -2$

when $c_1 \in (-2, 0)$, geometrically the optimal solution for (x_1, x_2) is $(\frac{14}{5}, \frac{4}{5})$

when $c_1 \in (-\infty, -2)$, geometrically the optimal solution for (x_1, x_2) is $(2, 0)$.

T2. According to the title, we can show a linear program:

assume x_1 is the number of bourbon whiskey and x_2 is blended whiskey, z is the profit of this project.

$$\text{maximize} \quad z = 6x_1(1-50\%) + 5.4x_2(1-30\%) - 5000$$

$$\text{subject to} \quad 3x_1 + 4x_2 \le 40000$$

$$5000 - 2x_1 + 6x_1 \cdot 50\% - 3x_2 + 5.4x_2 \cdot 30\% \ge 0.$$

$$x_1, x_2 \ge 0.$$

After simplization:

$$\text{maximize} \quad z = 3x_1 + 3.78x_2 - 5000$$

$$\text{s.t.} \quad 3x_1 + 4x_2 \le 40000$$

$$-x_1 + 1.38x_2 \le 5000$$

$$x_1, x_2 \ge 0.$$

T3. According to the problem, we can assume that x_i is the number of full-time consultants who works from time period i to $(i+1)$.

y_i is the number of part-time consultants who works at period i.

We can formulate the following LP model:

$$\text{minimize} \quad z = 8 \times 40 \times (x_1 + x_2 + x_3) + 4 \times 30 \times (y_1 + y_2 + y_3 + y_4)$$

$$\text{subject to} \quad x_1 + y_1 \ge 4$$

$$x_1 + x_2 + y_2 \ge 8$$

$$x_2 + x_3 + y_3 \ge 10$$

$$x_3 + y_4 \ge 6$$

$$x_1 \geq 2y_1$$

$$x_1 + x_2 \geq 2y_2$$

$$x_2 + x_3 \geq 2y_3$$

$$x_2 \geq 2y_4$$

$$x_1, x_2, x_3, y_1, y_2, y_3, y_4 \geq 0.$$

According to this model, we can satisfy the problem requirements.